# HAVING FIBRE DIMENSION D AT A POINT IS ETALE LOCAL ON THE SOURCE-AND-TARGET

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The definition on which the statement depends is delivered as a context span (role=context) rather than as a navigation stub, so the reader sees the notion of localness at a point being tested.

*Editorial note.* The counted spans carry 2 navigation stubs; no bibliography citation occurs in the spans.

*Editorial note.* The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

*Allowed standard prerequisites (editorial).*

- Definition of etale localness on the source-and-target for morphisms of germs. Same chapter, Definition definition-local-source-target-at-point, invoked at proof line 8293 and delivered as a context span. Supplies the shape of the diagram that must be checked.
- Existence of an etale morphism of fibres over the relevant points. Same chapter, Lemma lemma-etale-on-fiber, invoked at proof lines 8294-8296. Supplies the morphism of fibres through which the dimension is compared.
- Dimension at a point is invariant under an etale morphism of fibres. Same chapter, Lemma lemma-dimension-at-point-local, invoked at proof lines 8297-8298. Supplies the equality of the two dimensions.

*Source reference.* Original lemma 04N4: sources/stacks-descent/pinned-original-source.tex, source lines 5390–5390. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

*Source reference.* Original lemma 04NI: sources/stacks-descent/pinned-original-source.tex, source lines 8186–8186. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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*Date*: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Definition 0.1.** Let $\mathcal{Q}$ be a property of morphisms of germs of schemes. We say $\mathcal{Q}$ is *étale local on the source-and-target* if for any commutative diagram

$$\begin{array}{ccc} (U', u') & \xrightarrow[h']{} & (V', v') \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ (U, u) & \xrightarrow{h} & (V, v) \end{array}$$

of germs with étale vertical arrows we have $\mathcal{Q}(h) \Leftrightarrow \mathcal{Q}(h')$.

**Lemma 0.2.** *Let $d \in \{0, 1, 2, \ldots, \infty\}$. The property of morphisms of germs*

$$\mathcal{P}_d((X, x) \to (S, s)) \Leftrightarrow \dim_x(X_s) = d$$

*is étale local on the source-and-target.*

*Proof.* Given a diagram as in Definition 0.1 we obtain an étale morphism of fibres $U'_{v'} \to U_v$ mapping $u'$ to $u$, see Lemma 04NI. Hence now the equality $\dim_u(U_v) = \dim_{u'}(U'_{v'})$ follows from Lemma 04N4. □